

FPPF DESCENT AND LOCALIZING SUBCATEGORIES FOR ALGEBRAIC SPACES

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ABSTRACT. We prove descendability for fppf covers of quasi-compact quasi-separated algebraic spaces. Consequently, we show that point-generatedness is detectable along fppf covers. This allows us to classify $\otimes$ -localizing subcategories of the derived category of complexes with quasi-coherent cohomology on point-generated algebraic spaces in terms of subsets of its underlying topology.

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1. INTRODUCTION

1.1. What is known. Neeman classified localizing subcategories of the derived category of a Noetherian ring in terms of subsets of $\mathrm{Spec}(R)$ [Nee92, Theorem 2.8]. Every localizing subcategory is generated by residue fields at points. Beyond the Noetherian setting, this classification fails. See [Nee00].

The classification has been extended to Noetherian schemes. Alonso Tarrío–Jeremiás López–Souto Salorio established a bijection between subsets of a Noetherian scheme X and $\otimes$ -localizing subcategories of the derived category $D_{\mathrm{qc}}(X)$ of complexes with quasi-coherent cohomology [ATJLSSo4, Corollary 4.13]. There exist localizing subcategories of $D_{\mathrm{qc}}(\mathbb{P}_k^1)$ that are not $\otimes$ -compatible [ATJLSSo4, Example, pg. 595], so the $\otimes$ -condition is essential.

Stevenson [Ste13, Corollary 8.13] later recovered this classification using different methods. This is closely related to the tensor telescope conjecture. It asserts that on a rigidly compactly generated triangulated category, smashing $\otimes$ -Bousfield subcategories correspond to Thomason subsets of the Balmer spectrum. The connection is that smashing $\otimes$ -Bousfield subcategories are $\otimes$ -localizing subcategories.

Date: August 11, 2026.

2020 Mathematics Subject Classification. 14A30 (primary), 14A20, 13A35.

Key words and phrases. derived categories, algebraic spaces, localizing subcategories, tensor triangulated categories, point-generated.

The tensor telescope conjecture has been established in the case where D_{qc} is compactly generated [HR17]. For quasi-compact quasi-separated algebraic spaces, Hall–Rydh’s result applies because the Thomason condition holds, and it describes the relevant smashing $\otimes$ -Bousfield subcategories. However, this does not classify arbitrary $\otimes$ -localizing subcategories because a $\otimes$ -localizing subcategory need not be smashing. This work investigates classifications for $\otimes$ -localizing subcategories of D_{qc} on algebraic spaces.

1.2. What we do. In [AJL24], a quasi-compact quasi-separated scheme is called *point-generated* if D_{qc} coincides with the localizing subcategory generated by residue fields of points. This motivates the following:

Definition 1.1. A quasi-compact quasi-separated algebraic space X is called **point-generated** if there exists $S \subseteq |X|$ such that $D_{\text{qc}}(X) = \langle \{\mathbf{R}(t_s)_* \mathcal{O}_{\text{Spec}(k)}\} \rangle$ where t_s is a representative of $s \in S$.

The main observation of this paper is that point-generatedness satisfies a form of fppf ascent and descent. To prove this, we apply descendability techniques of Balmer and Mathew (see [Bal16, Proposition 3.15 & Definition 3.16] or [Mat16, Definition 3.18]). In particular, we show descendability for fppf covers of quasi-compact quasi-separated algebraic spaces.

Proposition 1.2. *Let $g: Y \rightarrow X$ be an fppf cover of quasi-compact quasi-separated algebraic spaces. Then $\mathbf{R}g_* \mathcal{O}_Y$ is a descendable algebra in $D_{\text{qc}}(X)$.*

Variations have been proved elsewhere. See e.g. [Hal22, Theorem 7.1] and [HLLP25, Theorem D]. However, we record a very simple proof in §3.

This allows us to prove that a point-generatedness of suitable algebraic space can be detected by its fppf covers.

Proposition 1.3. *Let $f: Y \rightarrow X$ be an fppf cover of quasi-compact quasi-separated algebraic spaces. Then X is point-generated if, and only if, Y is point-generated.*

As a consequence, there are many examples of point-generatedness.

Corollary 1.4. *Any Noetherian algebraic space X is point-generated.*

Proof. Choose an étale presentation $U \rightarrow X$ by an affine scheme. By [Nee92, Theorem 2.8] and [Sta26, Tag 071Q], U is point-generated. Thus, the claim follows from Proposition 1.3. $\square$

Combining this with the results of [AJL24] yields a classification:

Theorem 1.5. *Let X be a point-generated algebraic space. There exists a bijection*

$$\phi: \{ \otimes\text{-localizing subcategories of } D_{\text{qc}}(X) \} \longleftrightarrow \{ \text{subsets of } |X| \}: \theta$$

given as follows:

- $\phi(\mathcal{T})$ is the set of $p \in |X|$ represented by a $t: \text{Spec}(k) \rightarrow X$ such that $\mathbf{R}t_* \mathcal{O}_{\text{Spec}(k)} \in \mathcal{T}$
- θ sends a subset $S \subseteq |X|$ to the localizing subcategory generated by $\{\mathbf{R}(t_s)_* \mathcal{O}_{\text{Spec}(k)}\}_{s \in S}$, where t_s is a representative of $s \in S$.

These results show that $\otimes$ -localizing subcategories are generated by a set pushforwards of structure sheaves of spectra of fields. Our proofs do not mimic [AJL24]. Instead, we use descendability and étale presentations to reduce the problem to the scheme case.

Theorem 1.5 extends the classification of [AJL24] to algebraic spaces. In the Noetherian case, [AJL24, Theorem 0.1] can be replaced with [Nee92, Theorem 2.8], yielding an alternative proof that proceeds from affine schemes to Noetherian algebraic spaces.

Balchin–Omar Gómez–Stevenson have shown that the polynomial ring in countably many variables over a field is not point-generated [BOS26, Example 6.1]. **Proposition 1.3** implies the source of every fppf cover of this scheme is likewise not point-generated.

Acknowledgments. Lank was supported under the ERC Advanced Grant 101095900-TriCatApp. The author thanks Timothy De Deyn and Michal Hrbek for helpful discussions.

2. PRELIMINARIES

Here, ‘strictly full’ means a full subcategory closed under isomorphisms.

2.1. Conventions/notation. We follow [Sta26] for algebraic spaces and their derived categories. Let X be a quasi-compact quasi-separated algebraic space. $\text{Mod}(X)$ is the Grothendieck abelian category of sheaves of $\mathcal{O}_X$ -modules on the small étale site $X_{\text{étale}}$ of X [Sta26, Tag 03LT]. $\text{Qcoh}(X)$ is the full subcategory of $\text{Mod}(X)$ consisting of quasi-coherent sheaves. $D(X) := D(\text{Mod}(X))$ is the derived category of $\text{Mod}(X)$. $D_{\text{qc}}(X)$ is the full subcategory of $D(X)$ consisting of complexes with quasi-coherent cohomology sheaves. $\text{Perf}(X)$ is the full subcategory of perfect complexes in $D_{\text{qc}}(X)$.

2.2. Generation. We discuss generation for triangulated categories. See [BVdB03] for details. Let $\mathcal{T}$ be a triangulated category with shift functor $[1]: \mathcal{T} \rightarrow \mathcal{T}$. Consider a subcategory $\mathcal{S} \subseteq \mathcal{T}$. A triangulated subcategory of $\mathcal{T}$ is called **thick** if it is closed under direct summands. Denote by $\langle \mathcal{S} \rangle$ the smallest thick subcategory of $\mathcal{T}$ containing $\mathcal{S}$; if $\mathcal{S}$ consists of a single object G , we write $\langle G \rangle := \langle \mathcal{S} \rangle$. Set $\text{add}(\mathcal{S})$ to be the smallest strictly full subcategory of $\mathcal{T}$ containing $\mathcal{S}$ that is closed under shifts, finite coproducts, and direct summands. Inductively, let $\langle \mathcal{S} \rangle_0$ consist of all objects in $\mathcal{T}$ isomorphic to the zero object, $\langle \mathcal{S} \rangle_1 := \text{add}(\mathcal{S})$, and

$$\langle \mathcal{S} \rangle_n := \text{add}\{\text{cone}(\phi) \mid \phi \in \text{Hom}_{\mathcal{T}}(\langle \mathcal{S} \rangle_{n-1}, \langle \mathcal{S} \rangle_1)\}.$$

It can be checked that $\langle \mathcal{S} \rangle = \bigcup_{n=0}^{\infty} \langle \mathcal{S} \rangle_n$.

Suppose $\mathcal{T}$ admits small coproducts. Set $\text{Add}(\mathcal{S})$ to be the smallest strictly full subcategory of $\mathcal{T}$ containing $\mathcal{S}$ that is closed under shifts, small coproducts, and direct summands. Inductively, let $\overline{\langle \mathcal{S} \rangle}_0$ consist of all objects in $\mathcal{T}$ isomorphic to the zero object, $\overline{\langle \mathcal{S} \rangle}_1 := \text{Add}(\mathcal{S})$, and

$$\overline{\langle \mathcal{S} \rangle}_n := \text{Add}\{\text{cone}(\phi) \mid \phi \in \text{Hom}_{\mathcal{T}}(\overline{\langle \mathcal{S} \rangle}_{n-1}, \overline{\langle \mathcal{S} \rangle}_1)\}.$$

If $\mathcal{T}$ admits small coproducts, then the collection of compact objects in $\mathcal{T}$ is denoted by $\mathcal{T}^c$. These form a triangulated subcategory of $\mathcal{T}$. The **localizing subcategory** generated by a collection $\mathcal{S} \subseteq \mathcal{T}$, denoted by $\overline{\langle \mathcal{S} \rangle}$, is the smallest triangulated subcategory containing $\mathcal{S}$ and closed under small coproducts. An induction argument on n shows that $\langle \mathcal{S} \rangle_n \subseteq \overline{\langle \mathcal{S} \rangle}$ for all $n \geq 0$. If $\mathcal{T}^c$ is essentially small, then we say that $\mathcal{T}$ is **compactly generated** when $\mathcal{T} = \overline{\langle \mathcal{T}^c \rangle}$. Equivalently, $\mathcal{T}$ is compactly generated if, for any $E \in \mathcal{T}$ satisfying

$\mathrm{Hom}(P, E) = 0$ for all $P \in \mathcal{T}^c$, one has $E \cong 0$. Note that classical generators for $\mathcal{T}^c$ coincide with compact generators for $\mathcal{T}$ (see e.g. [Sta26, Tag 09SR]).

Assume that $\mathcal{T}$ is a tensor triangulated category with tensor $\otimes$ and unit 1 . The **localizing subcategory** generated by a collection $\mathcal{S} \subseteq \mathcal{T}$, denoted by $\overline{\langle \mathcal{S} \rangle}_{\otimes}$, is the smallest triangulated subcategory containing $\mathcal{S}$, closed under small coproducts, and the tensor action with respect to $\mathcal{T}$. A useful fact, used freely in our work, is that localizing subcategories in a triangulated category with small coproducts are closed under direct summands (see [BN93, Remark 1.4]).

2.3. Descendability. We recall a notion of descent for triangulated categories (see [Mat16, Definition 3.18] for the ∞ -categorical analog). Let $(\mathcal{T}, \otimes, 1)$ be a tensor triangulated category. Given $T \in \mathcal{T}$, denote by $\langle T \rangle_{\otimes}$ the smallest thick subcategory containing T that is closed under the tensor action by $\mathcal{T}$. A commutative monoid A in $\mathcal{T}$ is called **descendable**, if $\langle A \rangle_{\otimes} = \mathcal{T}$. See [DLM25, Lemma 6.3] for other characterizations.

Lemma 2.1. *Let $f: Y \rightarrow X$ be a morphism of quasi-compact quasi-separated algebraic spaces. Then $\langle \mathbf{R}f_* \mathcal{O}_Y \rangle_{\otimes} = \langle \mathbf{R}f_* D_{\mathrm{qc}}(Y) \rangle$. Consequently, $\mathbf{R}f_* \mathcal{O}_Y$ is descendable if, and only if, $\mathcal{O}_X \in \langle \mathbf{R}f_* D_{\mathrm{qc}}(Y) \rangle$.*

Proof. The second claim follows from the first by definition of descendability. We prove the first claim. Clearly, $\mathbf{R}f_* \mathcal{O}_Y \in \langle \mathbf{R}f_* D_{\mathrm{qc}}(Y) \rangle$. Moreover, an induction argument shows $\langle \mathbf{R}f_* \mathcal{O}_Y \rangle_{\otimes} \subseteq \langle \mathbf{R}f_* D_{\mathrm{qc}}(Y) \rangle$. Indeed, let $E \in \langle \mathbf{R}f_* D_{\mathrm{qc}}(Y) \rangle_n$. If $n = 0$, then $E \cong 0$, and so there exists nothing to prove. If $n = 1$, then E is a direct summand of a finite coproduct of the form $\oplus_{i \in I} \mathbf{R}f_* E_i$ for some $E_i \in D_{\mathrm{qc}}(Y)$, and projection formula can be used to check that $A \otimes^{\mathbf{L}} \oplus_{i \in I} \mathbf{R}f_* E_i \in \langle \mathbf{R}f_* D_{\mathrm{qc}}(Y) \rangle$ for all $A \in D_{\mathrm{qc}}(X)$. Assume we have proved the claim up to $n \geq 1$. Let $E \in \langle \mathbf{R}f_* D_{\mathrm{qc}}(Y) \rangle_{n+1}$. There exists $A \in \langle \mathbf{R}f_* D_{\mathrm{qc}}(Y) \rangle_n$, $B \in \langle \mathbf{R}f_* D_{\mathrm{qc}}(Y) \rangle_1$, and a distinguished triangle

$$A \rightarrow E \oplus E' \rightarrow B \rightarrow A[1].$$

Choose any $G \in D_{\mathrm{qc}}(X)$. Tensoring with G yields a distinguished triangle

$$G \otimes^{\mathbf{L}} A \rightarrow G \otimes^{\mathbf{L}} (E \oplus E') \rightarrow G \otimes^{\mathbf{L}} B \rightarrow (G \otimes^{\mathbf{L}} A)[1].$$

The induction hypothesis says $G \otimes^{\mathbf{L}} A, G \otimes^{\mathbf{L}} B \in \langle \mathbf{R}f_* D_{\mathrm{qc}}(Y) \rangle$, and so,

$$(G \otimes^{\mathbf{L}} E) \oplus (G \otimes^{\mathbf{L}} E') \cong G \otimes^{\mathbf{L}} (E \oplus E') \in \langle \mathbf{R}f_* D_{\mathrm{qc}}(Y) \rangle.$$

To check the reverse inclusion, it suffices to show $\mathbf{R}f_* D_{\mathrm{qc}}(Y) \subseteq \langle \mathbf{R}f_* \mathcal{O}_Y \rangle_{\otimes}$. Choose $E \in D_{\mathrm{qc}}(Y)$. Then unit equations for adjunction yield a composition

$$\mathbf{R}f_* E \rightarrow \mathbf{R}f_* \mathbf{L}f^* \mathbf{R}f_* E \rightarrow \mathbf{R}f_* E$$

which is the identity on $\mathbf{R}f_* E$. Hence, $\mathbf{R}f_* E \in \langle \mathbf{R}f_* \mathbf{L}f^* \mathbf{R}f_* E \rangle_1$. Projection formula says that

$$\mathbf{R}f_* \mathbf{L}f^* \mathbf{R}f_* E \cong \mathbf{R}f_* E \otimes^{\mathbf{L}} \mathbf{R}f_* \mathcal{O}_Y.$$

Since $\mathbf{R}f_* E \otimes^{\mathbf{L}} \mathbf{R}f_* \mathcal{O}_Y \in \langle \mathbf{R}f_* \mathcal{O}_Y \rangle_{\otimes}$, the claim follows. $\square$

3. DESCENT

We prove a version of descent.

Lemma 3.1. *Let $f: Y \rightarrow X$ be an fppf cover of quasi-compact quasi-separated algebraic spaces. If X is a scheme, then $\mathbf{R}f_*\mathcal{O}_Y$ is a descendable algebra in $D_{\text{qc}}(X)$.*

Proof. Choose an étale presentation $s: U \rightarrow Y$ from an affine scheme. By [Sta26, Tag 03Y9], $f \circ s$ is an fppf cover. Moreover, by [BS17, Lemma 11.23(2)], $\mathbf{R}(f \circ s)_*\mathcal{O}_U$ is descendable. Recall that there exists a natural isomorphism $\mathbf{R}f_* \circ \mathbf{R}s_* \rightarrow \mathbf{R}(f \circ s)_*$ [Sta26, Tag 0D6E]. Thus, from Lemma 2.1, it follows that

$$\mathcal{O}_X \in \langle \mathbf{R}(f \circ s)_*D_{\text{qc}}(U) \rangle \subseteq \langle \mathbf{R}f_*D_{\text{qc}}(Y) \rangle,$$

which yields that $\mathbf{R}f_*\mathcal{O}_Y$ is a descendable algebra in $D_{\text{qc}}(X)$. $\square$

Proof of Proposition 1.2. Let P be the property for quasi-compact open immersions $U \rightarrow X$ that $\mathcal{O}_U \in \langle \mathbf{R}(g_U)_*D_{\text{qc}}(U \times_X Y) \rangle$ where $g_U: Y_U := U \times_X Y \rightarrow U$ is the base change of g along $U \rightarrow X$. We apply [Sta26, Tag 09IT]. We may assume X is not empty. By [Sta26, Tag 06NH], X contains a nonempty subspace X_0 which is a scheme. Choose a nonempty affine open subscheme W_0 of X_0 . Then W_0 satisfies the desired claim via Lemma 3.1.

We check the remaining condition in [Sta26, Tag 09IT]. Consider an elementary distinguished square given by a quasi-compact open immersion $j: U \rightarrow W$ such that $W_0 \subseteq U$, an étale morphism $f: X' \rightarrow W$ from an affine scheme, and W an open subspace of X . Assume that U satisfies the desired claim. Form the fibered square

$$\begin{array}{ccc} U' & \xrightarrow{j'} & X' \\ f' \downarrow & & \downarrow f \\ U & \xrightarrow{j} & W. \end{array}$$

By [Sta26, Tag 08GW], there exists a distinguished triangle

$$\mathcal{O}_W \rightarrow \mathbf{R}j_*\mathcal{O}_U \oplus \mathbf{R}f_*\mathcal{O}_{X'} \rightarrow \mathbf{R}(f \circ j')_*\mathcal{O}_{U'} \rightarrow \mathcal{O}_W[1].$$

Consider the commutative cube

$$\begin{array}{ccccc} Y'_U & \xrightarrow{j'_Y} & Y' & & \\ \downarrow g' & \searrow & \downarrow g_{X'} & & \\ U' & \xrightarrow{j'} & X' & & \\ \downarrow f' & \searrow f'_Y & \downarrow f & & \\ & & Y_U & \xrightarrow{\tilde{j}} & Y_W \\ & & \downarrow g_U & & \downarrow g_W \\ & & U & \xrightarrow{j} & W \end{array}$$

whose faces are fibered. By [Sta26, Tag 03Y9], g_U , $g_{X'}$, and g' are fppf covers. Since U satisfies property P , it follows that $\mathcal{O}_U \in \langle \mathbf{R}(g_U)_*D_{\text{qc}}(Y_U) \rangle$. Moreover, Lemma 3.1 implies $\mathcal{O}_{X'} \in \langle \mathbf{R}(g_{X'})_*D_{\text{qc}}(Y') \rangle$ and $\mathcal{O}_{U'} \in \langle \mathbf{R}(g')_*D_{\text{qc}}(Y'_U) \rangle$. See Lemma 2.1. Consequently, we

obtain

$$\begin{aligned}
\mathcal{O}_W &\in \langle \mathbf{R}j_*\mathcal{O}_U \oplus \mathbf{R}f_*\mathcal{O}_{X'} \oplus \mathbf{R}(f \circ j')_*\mathcal{O}_{U'} \rangle \\
&\subseteq \langle \mathbf{R}(j \circ g_U)_*D_{\text{qc}}(Y_U) \oplus \mathbf{R}(f \circ g_{X'})_*D_{\text{qc}}(Y') \oplus \mathbf{R}(j \circ f' \circ g')_*D_{\text{qc}}(Y'_U) \rangle \\
&\subseteq \langle \mathbf{R}(g_W \circ \tilde{j})_*D_{\text{qc}}(Y_U) \oplus \mathbf{R}(g_W \circ \tilde{f})_*D_{\text{qc}}(Y') \oplus \mathbf{R}(g_W \circ \tilde{j} \circ f'_Y)_*D_{\text{qc}}(Y'_U) \rangle \\
&\subseteq \langle \mathbf{R}(g_W)_*D_{\text{qc}}(Y_W) \rangle.
\end{aligned}$$

By Lemma 2.1, this completes the proof. $\square$

4. LOCALIZING

Lemma 4.1. *Let $F: \mathcal{T} \rightarrow \mathcal{S}$ be an exact functor of triangulated categories admitting small coproducts. Suppose $\mathcal{A} \subseteq \mathcal{T}$, that F preserves small coproducts, and $\mathcal{B} \subseteq \mathcal{S}$ is localizing. Then $F^{-1}\mathcal{B} \subseteq \mathcal{T}$ is localizing and $F(\langle \mathcal{A} \rangle) \subseteq \overline{\langle F(\mathcal{A}) \rangle}$.*

Proof. This is well-known, but we include it for convenience. See e.g. [HR17, Lemma 1.1]. We prove the first claim. Recall that $F^{-1}\mathcal{B} \subseteq \mathcal{T}$ denotes the strictly full subcategory of $E \in \mathcal{T}$ such that $F(E) \in \mathcal{B}$. If $E \in F^{-1}\mathcal{B}$ and $n \in \mathbb{Z}$, then $F(E[n]) \cong F(E)[n] \in \mathcal{B}$ because $\mathcal{B}$ is closed under shifts, and so $E[n] \in F^{-1}\mathcal{B}$. Moreover, for any distinguished triangle

$$A \rightarrow E \rightarrow B \rightarrow A[1]$$

where $A, B \in F^{-1}\mathcal{B}$, it follows that

$$F(A) \rightarrow F(E) \rightarrow F(B) \rightarrow F(A[1])$$

is a distinguished triangle of objects in $\mathcal{B}$. Hence, $E \in F^{-1}\mathcal{B}$, which shows that $F^{-1}\mathcal{B}$ is a triangulated subcategory of $\mathcal{T}$. Finally, if I is a set and $E_i \in F^{-1}\mathcal{B}$ for $i \in I$, then $F(\oplus_{i \in I} E_i) \cong \oplus_{i \in I} F(E_i) \in \mathcal{B}$ because $\mathcal{B}$ is closed under small coproducts and F preserves small coproducts. This implies that $\oplus_{i \in I} E_i \in F^{-1}\mathcal{B}$.

We prove the second claim. Since $F(\mathcal{A}) \subseteq \overline{\langle F(\mathcal{A}) \rangle}$, it follows that $\overline{\langle F(\mathcal{A}) \rangle} \subseteq \overline{\langle F(\langle \mathcal{A} \rangle) \rangle}$. By the first claim, we know that $F^{-1}\overline{\langle F(\mathcal{A}) \rangle}$ is localizing. As $\mathcal{A} \subseteq F^{-1}\overline{\langle F(\mathcal{A}) \rangle}$, we obtain $\langle \mathcal{A} \rangle \subseteq F^{-1}\overline{\langle F(\mathcal{A}) \rangle}$. Therefore, the desired claim follows. $\square$

Lemma 4.2. *Let X be a quasi-compact quasi-separated algebraic space, $S \subseteq |X|$, and fix a representative $t_s: \text{Spec}(k) \rightarrow X$ for each $s \in S$. Then*

$$\overline{\langle \{\mathbf{R}(t_s)_*\mathcal{O}_{\text{Spec}(k)}\}_{s \in S} \rangle}_{\otimes} = \overline{\langle \{\mathbf{R}(t_s)_*\mathcal{O}_{\text{Spec}(k)}\}_{s \in S} \rangle}.$$

Moreover, this localizing subcategory is independent of the choice of representatives for $s \in S$.

Proof. We prove the first claim. It is easy to see that

$$\overline{\langle \{\mathbf{R}(t_s)_*\mathcal{O}_{\text{Spec}(k)}\}_{s \in S} \rangle} \subseteq \overline{\langle \{\mathbf{R}(t_s)_*\mathcal{O}_{\text{Spec}(k)}\}_{s \in S} \rangle}_{\otimes}.$$

To prove the reverse inclusion, we show that $\overline{\langle \{\mathbf{R}(t_s)_*\mathcal{O}_{\text{Spec}(k)}\}_{s \in S} \rangle}$ is a $\otimes$ -subcategory. Let $E \in D_{\text{qc}}(X)$. Note that $E \otimes^{\mathbf{L}} (-)$ is a small coproduct preserving endofunctor on $D_{\text{qc}}(X)$. By the projection formula and Lemma 4.1,

$$\begin{aligned}
E \otimes^{\mathbf{L}} \overline{\langle \{\mathbf{R}(t_s)_*\mathcal{O}_{\text{Spec}(k)}\}_{s \in S} \rangle} &\subseteq \overline{\langle E \otimes^{\mathbf{L}} \{\mathbf{R}(t_s)_*\mathcal{O}_{\text{Spec}(k)}\}_{s \in S} \rangle} \\
&\subseteq \overline{\langle \{E \otimes^{\mathbf{L}} \mathbf{R}(t_s)_*\mathcal{O}_{\text{Spec}(k)}\}_{s \in S} \rangle} \\
&\subseteq \overline{\langle \{\mathbf{R}(t_s)_*\mathbf{L}t_s^*E\}_{s \in S} \rangle}.
\end{aligned}$$

Moreover, for each $s \in S$, $D_{\text{qc}}(\text{Spec}(k)) = \overline{\langle \mathcal{O}_{\text{Spec}(k)} \rangle}$. By [Lemma 4.1](#),

$$\overline{\langle \mathbf{R}(t_s)_* \mathbf{L}t_s^* E \rangle} \subseteq \overline{\langle \mathbf{R}(t_s)_* \mathcal{O}_{\text{Spec}(k)} \rangle}.$$

Consequently, it follows that

$$\begin{aligned} E \otimes^{\mathbf{L}} \overline{\langle \{\mathbf{R}(t_s)_* \mathcal{O}_{\text{Spec}(k)}\}_{s \in S} \rangle} &\subseteq \overline{\langle \{\mathbf{R}(t_s)_* \mathbf{L}t_s^* E\}_{s \in S} \rangle} \\ &\subseteq \overline{\langle \{\mathbf{R}(t_s)_* \mathcal{O}_{\text{Spec}(k)}\}_{s \in S} \rangle}. \end{aligned}$$

Next we check the second claim. Choose any collection of representatives $v_s: \text{Spec}(k') \rightarrow X$ for each $s \in S$. Since t_s and v_s represent s , there exist a field ℓ and commutative diagram

$$\begin{array}{ccc} \text{Spec}(\ell) & \xrightarrow{t'_s} & \text{Spec}(k') \\ v'_s \downarrow & & \downarrow v_s \\ \text{Spec}(k) & \xrightarrow{t_s} & X. \end{array}$$

Note that t'_s and v'_s are faithfully flat. Hence, the derived pushforward of $\mathcal{O}_{\text{Spec}(\ell)}$ is nonzero along each morphism. In particular, $\mathbf{R}(t'_s)_* \mathcal{O}_{\text{Spec}(\ell)}$ is a nonzero coproduct of shifts of $\mathcal{O}_{\text{Spec}(k')}$, and similarly for $\mathbf{R}(v'_s)_* \mathcal{O}_{\text{Spec}(\ell)}$. Consequently,

$$\overline{\langle \mathbf{R}(t'_s)_* \mathcal{O}_{\text{Spec}(\ell)} \rangle} = D_{\text{qc}}(\text{Spec}(k')) = \overline{\langle \mathcal{O}_{\text{Spec}(k')} \rangle}.$$

A similar statement holds for $\mathbf{R}(v'_s)_* \mathcal{O}_{\text{Spec}(\ell)}$ in $D_{\text{qc}}(\text{Spec}(k))$. By [Lemma 4.1](#), we have that

$$\begin{aligned} \mathbf{R}(v_s)_* \overline{\langle \mathbf{R}(t'_s)_* \mathcal{O}_{\text{Spec}(\ell)} \rangle} &\subseteq \overline{\langle \mathbf{R}(v_s)_* \mathcal{O}_{\text{Spec}(k')} \rangle}, \\ \mathbf{R}(v_s)_* \overline{\langle \mathcal{O}_{\text{Spec}(k')} \rangle} &\subseteq \overline{\langle \mathbf{R}(v_s)_* \mathbf{R}(t'_s)_* \mathcal{O}_{\text{Spec}(\ell)} \rangle}. \end{aligned}$$

This implies that

$$\overline{\langle \mathbf{R}(v_s)_* \mathcal{O}_{\text{Spec}(k')} \rangle} = \overline{\langle \mathbf{R}(v_s)_* \mathbf{R}(t'_s)_* \mathcal{O}_{\text{Spec}(\ell)} \rangle}.$$

A similar statement holds for $\mathbf{R}(v'_s)_* \mathcal{O}_{\text{Spec}(\ell)}$. Therefore, it follows that

$$\overline{\langle \mathbf{R}(v_s)_* \mathcal{O}_{\text{Spec}(k')} \rangle} = \overline{\langle \mathbf{R}(v_s)_* \mathbf{R}(t'_s)_* \mathcal{O}_{\text{Spec}(\ell)} \rangle} = \overline{\langle \mathbf{R}(t_s)_* \mathcal{O}_{\text{Spec}(k)} \rangle},$$

which completes the proof. $\square$

Lemma 4.3. *Let X be a quasi-compact quasi-separated algebraic space. Suppose G is a compact generator for $D_{\text{qc}}(X)$. If $\mathcal{S} \subseteq D_{\text{qc}}(X)$ is a set, then $\overline{\langle \mathcal{S} \rangle}_{\otimes} = \overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle}$.*

Proof. It is easy to see that $\overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle} \subseteq \overline{\langle \mathcal{S} \rangle}_{\otimes}$. We prove the reverse inclusion. We show that $\overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle}$ is an $\otimes$ -subcategory. By [\[Sta26, Tag 09SN\]](#), every $E \in D_{\text{qc}}(X)$ admits a distinguished triangle

$$\oplus_{n \geq 1} P_n \rightarrow \oplus_{n \geq 1} P_n \rightarrow E \rightarrow (\oplus_{n \geq 1} P_n)[1]$$

where $P_n \in \overline{\langle G \rangle}_n$. To show that $E \otimes^{\mathbf{L}} \overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle} \subseteq \overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle}$, it suffices to prove that $\oplus_{n \geq 1} P_n \otimes^{\mathbf{L}} \overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle} \subseteq \overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle}$.

As localizing subcategories are closed under small coproducts, we can reduce to proving that each $P_n \otimes^{\mathbf{L}} \overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle} \subseteq \overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle}$. However, $P_n \in \overline{\langle G \rangle}_n$, and so it suffices to verify that $\overline{\langle G \rangle}_n \otimes^{\mathbf{L}} \overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle} \subseteq \overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle}$. The claim now follows by induction on n . Indeed, let

$A \in \overline{\langle G \rangle}_1$. Then A is a direct summand of a small coproduct of shifts of G , say $\bigoplus_{i \in I} G[i]$. So, for any $S \in \mathcal{S}$,

$$\left(\bigoplus_{i \in I} G[i] \right) \otimes^{\mathbf{L}} (G \otimes^{\mathbf{L}} S) \cong \bigoplus_{i \in I} (G[i] \otimes^{\mathbf{L}} G \otimes^{\mathbf{L}} S).$$

Since $G[i] \otimes^{\mathbf{L}} G \in \langle G \rangle$, it follows that each $G[i] \otimes^{\mathbf{L}} G \otimes^{\mathbf{L}} S \in \overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle}$. By Lemma 4.1,

$$\left(\bigoplus_{i \in I} G[i] \right) \otimes^{\mathbf{L}} \overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle} \subseteq \overline{\left(\bigoplus_{i \in I} G[i] \right) \otimes^{\mathbf{L}} G \otimes^{\mathbf{L}} \mathcal{S}} \subseteq \overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle}.$$

Now, let $E \in \overline{\langle G \rangle}_{n+1}$. There exists a distinguished triangle

$$A \rightarrow E \oplus E' \rightarrow B \rightarrow A[1]$$

where $A \in \overline{\langle G \rangle}_n$ and $B \in \overline{\langle G \rangle}_1$. Choose any $Q \in \overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle}$. Tensoring with Q gives a distinguished triangle

$$A \otimes^{\mathbf{L}} Q \rightarrow E \otimes^{\mathbf{L}} Q \oplus E' \otimes^{\mathbf{L}} Q \rightarrow B \otimes^{\mathbf{L}} Q \rightarrow A[1] \otimes^{\mathbf{L}} Q$$

By the induction hypothesis, it follows that $A \otimes^{\mathbf{L}} Q, B \otimes^{\mathbf{L}} Q \in \overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle}$. Hence, $E \otimes^{\mathbf{L}} Q \in \overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle}$.

Since $\mathcal{O}_X \in \langle G \rangle$, it follows that for each $E \in \mathcal{S}$,

$$E \in \langle G \rangle \otimes^{\mathbf{L}} E \subseteq \langle G \otimes^{\mathbf{L}} E \rangle \subseteq \overline{\langle G \otimes^{\mathbf{L}} E \rangle}.$$

Thus, $\overline{\langle G \otimes^{\mathbf{L}} \mathcal{S} \rangle}$ is a $\otimes$ -subcategory which contains $\mathcal{S}$. $\square$

Lemma 4.4. *Let $\mathcal{T}$ be a triangulated category and $\mathcal{S} \subseteq \mathcal{T}$ be a subcategory. For each $E \in \langle \mathcal{S} \rangle_n$, there exists an $S \in \langle \mathcal{S} \rangle_1$ with $E \in \langle S \rangle_n$.*

Proof. This is essentially [DLM25, Lemma 5.5] but we rephrase it a bit. There is nothing to prove when $E = 0$, and so, we can assume that $E \neq 0$. In this case, $n > 0$. If $n = 1$, then $S = E$ satisfies the desired claim. Assume the desired claim holds for all $A \in \langle \mathcal{S} \rangle_c$ with $0 \leq c \leq n$. Let $E \in \langle \mathcal{S} \rangle_{n+1}$. This gives us a distinguished triangle

$$A \rightarrow E \oplus E' \rightarrow B \rightarrow A[1]$$

with $A \in \langle \mathcal{S} \rangle_n$ and $B \in \langle \mathcal{S} \rangle_1$. By the induction hypothesis, there exists $S' \in \langle \mathcal{S} \rangle_1$ with $A \in \langle S' \rangle_n$. Define $S := S' \oplus B$. Clearly, $S \in \langle \mathcal{S} \rangle_1$, and so the distinguished triangle above shows $E \in \langle S \rangle_{n+1}$. This completes the proof. $\square$

5. FPPF COVERS

Lemma 5.1. *Let X be a quasi-compact quasi-separated algebraic space. Assume there exists an fpf cover $s: U \rightarrow X$ from a point-generated affine scheme. If $\mathcal{T}$ is a $\otimes$ -localizing subcategory of $D_{\text{qc}}(X)$, then there exists $S \subseteq |X|$ such that $\mathcal{T} = \overline{\langle \{ \mathbf{R}(t_s)_* \mathcal{O}_{\text{Spec}(k)} \}_{s \in S} \rangle}$ where $t_s: \text{Spec}(k) \rightarrow X$ is a representative for each $s \in S$. In particular, X is point-generated.*

Proof. By Proposition 1.2, there exists an $n \geq 0$ such that $\mathcal{O}_X \in \langle \mathbf{R}s_* D_{\text{qc}}(U) \rangle_n$. By Lemma 4.4, there exists $E \in D_{\text{qc}}(U)$ such that $\mathcal{O}_X \in \langle \mathbf{R}s_* E \rangle_n$. Indeed, there exists $E' \in \langle \mathbf{R}s_* D_{\text{qc}}(U) \rangle_1$ such that $\mathcal{O}_X \in \langle E' \rangle_n$. However, E' is a direct summand of a finite coproduct of the form $\bigoplus_{i \in I} \mathbf{R}s_* E_i$. Taking $E := \bigoplus_{i \in I} E_i$ and using that $\mathbf{R}s_*$ commutes with small coproducts shows that $\mathcal{O}_X \in \langle \mathbf{R}s_* E \rangle_n$.

Fix a $\otimes$ -localizing subcategory $\mathcal{T} \subseteq D_{\text{qc}}(X)$. By [Lemma 4.1](#), $s^{-1}\mathcal{T}$ is localizing in $D_{\text{qc}}(U)$. Since U is affine, $s^{-1}\mathcal{T}$ is $\otimes$ -localizing because every localizing subcategory is $\otimes$ -closed (e.g. use that $\mathcal{O}_U$ is a compact generator). Denote by $h_q: \text{Spec}(\kappa(q)) \rightarrow U$ the natural morphisms for each $q \in |U|$. By [\[Sta26, Tags 03BV & 03BU\]](#), the points of U (viewed as a topological space) are in one-to-one correspondence with those of $|U|$. By [\[AJL24, Theorem 0.1\]](#) and [\[Sta26, Tag 071Q\]](#), there exists $T \subseteq |U|$ such that

$$\overline{\langle \{\mathbf{R}(h_q)_* \mathcal{O}_{\text{Spec}(\kappa(q))}\}_{q \in T} \rangle} = s^{-1}\mathcal{T}.$$

Note that the construction of $s^{-1}\mathcal{T}$ implies

$$\overline{\langle \{\mathbf{R}(s \circ h_q)_* \mathcal{O}_{\text{Spec}(\kappa(q))}\}_{q \in T} \rangle} \subseteq \mathcal{T}.$$

We prove the reverse inclusion. Let $A \in \mathcal{T}$. By tensoring with A , we obtain $A \in \langle \mathbf{R}s_* E \otimes^{\mathbf{L}} A \rangle_n$. By the projection formula, $\mathbf{R}s_* E \otimes^{\mathbf{L}} A \cong \mathbf{R}s_*(E \otimes^{\mathbf{L}} \mathbf{L}s^* A)$. Since $A \in \mathcal{T}$ and $\mathcal{T}$ is $\otimes$ -localizing, we know that $\mathbf{R}s_* E \otimes^{\mathbf{L}} A \in \mathcal{T}$. Hence, $E \otimes^{\mathbf{L}} \mathbf{L}s^* A \in s^{-1}\mathcal{T}$, which implies

$$\mathbf{R}s_*(E \otimes^{\mathbf{L}} \mathbf{L}s^* A) \in \mathbf{R}s_* s^{-1}\mathcal{T}.$$

Applying [Lemma 4.1](#),

$$\mathbf{R}s_* s^{-1}\mathcal{T} = \mathbf{R}s_* \overline{\langle \{\mathbf{R}(h_q)_* \mathcal{O}_{\text{Spec}(\kappa(q))}\}_{q \in T} \rangle} \subseteq \overline{\langle \{\mathbf{R}(s \circ h_q)_* \mathcal{O}_{\text{Spec}(\kappa(q))}\}_{q \in T} \rangle}.$$

Thus,

$$A \in \langle \mathbf{R}s_* E \otimes^{\mathbf{L}} A \rangle_n \subseteq \overline{\langle \{\mathbf{R}(s \circ h_q)_* \mathcal{O}_{\text{Spec}(\kappa(q))}\}_{q \in T} \rangle}.$$

By [Lemma 4.2](#), the choice of representatives for $s(T) \subseteq |X|$ does not matter. $\square$

Lemma 5.2. *Let $f: Y \rightarrow X$ be an fppf cover of quasi-compact quasi-separated algebraic spaces. If X is point-generated, then Y is point-generated.*

Proof. Choose an étale presentation $s: U \rightarrow Y$. By [Lemma 5.1](#), it suffices to show that U is point-generated. By [Proposition 1.2](#) and [Lemma 2.1](#), $\mathcal{O}_X \in \langle \mathbf{R}(f \circ s)_* D_{\text{qc}}(U) \rangle$. In fact, there exists $n \geq 0$ such that $\mathcal{O}_X \in \langle \mathbf{R}(f \circ s)_* D_{\text{qc}}(U) \rangle_n$. Applying $\mathbf{L}(f \circ s)^*$, it follows that

$$\mathcal{O}_U \in \langle \mathbf{L}(f \circ s)^* \mathbf{R}(f \circ s)_* D_{\text{qc}}(U) \rangle_n \subseteq \langle \mathbf{L}(f \circ s)^* D_{\text{qc}}(X) \rangle_n.$$

Since X is point-generated, there exists $S \subseteq |X|$ such that $D_{\text{qc}}(X) = \overline{\langle \{\mathbf{R}(t_s)_* \mathcal{O}_{\text{Spec}(k)}\}_{s \in S} \rangle}$ where t_s is some representative of $s \in S$. As $\mathbf{L}(f \circ s)^*$ preserves small coproducts, [Lemma 4.1](#) implies

$$\mathbf{L}(f \circ s)^* \overline{\langle \{\mathbf{R}(t_s)_* \mathcal{O}_{\text{Spec}(k)}\}_{s \in S} \rangle} \subseteq \overline{\langle \{\mathbf{L}(f \circ s)^* \mathbf{R}(t_s)_* \mathcal{O}_{\text{Spec}(k)}\}_{s \in S} \rangle}$$

Now, a localizing subcategory is triangulated, so it is closed under extensions and shifts, which implies

$$\mathcal{O}_U \in \langle \mathbf{L}(f \circ s)^* D_{\text{qc}}(X) \rangle_n \subseteq \overline{\langle \{\mathbf{L}(f \circ s)^* \mathbf{R}(t_s)_* \mathcal{O}_{\text{Spec}(k)}\}_{s \in S} \rangle}.$$

Fix t_s . Consider the fibered square

$$\begin{array}{ccc} U \times_X \text{Spec}(k) & \xrightarrow{p_2^s} & \text{Spec}(k) \\ p_1^s \downarrow & & \downarrow t_s \\ U & \xrightarrow{f \circ s} & X. \end{array}$$

Base change says p_2^s is an fppf cover. By [Sta26, Tag 09TF], t_s is affine. Again, by base change, p_1^s is affine. Hence, $U \times_X \text{Spec}(k)$ is affine, and so $U \times_X \text{Spec}(k)$ is finitely presented over k . This ensures that $U \times_X \text{Spec}(k)$ is Noetherian. By [Neeg2, Theorem 2.8] and [Sta26, Tag 071Q], $U \times_X \text{Spec}(k)$ is point-generated, and so,

$$\overline{\langle \mathbb{O}_{U \times_X \text{Spec}(k)} \rangle} = \overline{\langle \{ \mathbf{R}(v_l)_* \mathbb{O}_{\text{Spec}(\kappa(l))} \}_{l \in |U \times_X \text{Spec}(k)|} \rangle}.$$

Flat base change implies

$$\mathbf{R}(p_1^s)_* \mathbb{O}_{U \times_X \text{Spec}(k)} \cong \mathbf{L}(f \circ s)^* \mathbf{R}(t_s)_* \mathbb{O}_{\text{Spec}(k)}.$$

Since $\mathbf{R}(p_1^s)_*$ admits a right adjoint [Sta26, Tag 0E55] it preserves small coproducts. Thus, Lemma 4.1 implies

$$\mathbf{R}(p_1^s)_* \overline{\langle \mathbb{O}_{U \times_X \text{Spec}(k)} \rangle} \subseteq \overline{\langle \{ \mathbf{R}(p_1^s \circ v_l)_* \mathbb{O}_{\text{Spec}(\kappa(l))} \}_{l \in |U \times_X \text{Spec}(k)|} \rangle}.$$

Flat base change implies

$$\mathbf{R}(p_1^s)_* \mathbb{O}_{U \times_X \text{Spec}(k)} \cong \mathbf{L}(f \circ s)^* \mathbf{R}(t_s)_* \mathbb{O}_{\text{Spec}(k)}.$$

Consequently, we obtain that

$$\begin{aligned} \mathbb{O}_U &\in \langle \mathbf{L}(f \circ s)^* D_{\text{qc}}(X) \rangle_n \\ &\subseteq \overline{\langle \{ \mathbf{L}(f \circ s)^* \mathbf{R}(t_s)_* \mathbb{O}_{\text{Spec}(k)} \}_{s \in S} \rangle} \\ &\subseteq \overline{\langle \{ \{ \mathbf{R}(p_1^s \circ v_l)_* \mathbb{O}_{\text{Spec}(\kappa(l))} \}_{l \in |U \times_X \text{Spec}(k)|} \}_{s \in S} \rangle}. \end{aligned}$$

However, Lemma 4.2 says the latter category above is closed under tensoring by objects from $D_{\text{qc}}(U)$. Therefore, since this latter category contains the tensor unit $\mathbb{O}_U$, it follows that U is point-generated. $\square$

Proof of Proposition 1.3. This follows from Lemmas 5.1 and 5.2. $\square$

Proof of Theorem 1.5. Choose an étale presentation $s : U \rightarrow X$ with U affine. In particular, s is an fppf cover. Since X is point-generated, Proposition 1.3 implies that U is point-generated. If $u \in |U|$, let $h_u : \text{Spec}(\kappa(u)) \rightarrow U$ be the natural morphism.

Lemma 4.2 implies that whether $\mathbf{R}t_* \mathbb{O}_{\text{Spec}(k)}$ belongs to a $\otimes$ -localizing subcategory depends only on the point represented by $t : \text{Spec}(k) \rightarrow X$. The same result shows that $\theta(S)$ is independent of the chosen representatives and is $\otimes$ -localizing. Thus, both ϕ and θ are well-defined.

We first prove that $\theta \circ \phi$ is the identity. Let $\mathcal{T} \subseteq D_{\text{qc}}(X)$ be a $\otimes$ -localizing subcategory. By Lemma 5.1, there exists a subset $S \subseteq |X|$ such that $\mathcal{T} = \theta(S)$. Since every point of S is represented by $t : \text{Spec}(k) \rightarrow X$ such that $\mathbf{R}t_* \mathbb{O}_{\text{Spec}(k)}$ belonging to $\mathcal{T}$, we have $S \subseteq \phi(\mathcal{T})$. Consequently, $\mathcal{T} = \theta(S) \subseteq \theta(\phi(\mathcal{T}))$.

Conversely, let $p \in \phi(\mathcal{T})$. By the definition of ϕ , there is a representative $t : \text{Spec}(k) \rightarrow X$ of p such that $\mathbf{R}t_* \mathbb{O}_{\text{Spec}(k)} \in \mathcal{T}$. By Lemma 4.2, the point object associated to any other representative of p also belongs to $\mathcal{T}$. It follows that every generator of $\theta(\phi(\mathcal{T}))$ belongs to $\mathcal{T}$, and hence $\theta(\phi(\mathcal{T})) \subseteq \mathcal{T}$. Therefore, $\theta \circ \phi = 1$.

We next prove that $\phi \circ \theta$ is the identity. Fix a subset $S \subseteq |X|$. For every $p \in |S|$, choose a point $r_p \in |U|$ such that $s(r_p) = p$. Set $t_p := s \circ h_{r_p} : \text{Spec}(\kappa(r_p)) \rightarrow X$. The morphism

t_p represents p [GL26, Lemma 2.4], so Lemma 4.2 allows us to compute $\theta(S)$ using these representatives:

$$\theta(S) = \overline{\langle \{\mathbf{R}(t_p)_* \mathcal{O}_{\mathrm{Spec}(\kappa(r_p))}\}_{p \in |S|} \rangle}.$$

It is immediate that $S \subseteq \phi(\theta(S))$. We are left to prove the reverse inclusion.

Let $q \in \phi(\theta(S))$. Since s is surjective, choose $r \in |U|$ such that $s(r) = q$. Put $t_r := s \circ h_r: \mathrm{Spec}(\kappa(r)) \rightarrow X$. Since t_r represents q , the well-definedness of ϕ gives $\mathbf{R}(t_r)_* \mathcal{O}_{\mathrm{Spec}(\kappa(r))} \in \theta(S)$. Applying $\mathbf{L}s^*$ and Lemma 4.1, we obtain

$$(5.1) \quad \mathbf{L}s^* \mathbf{R}(t_r)_* \mathcal{O}_{\mathrm{Spec}(\kappa(r))} \in \overline{\langle \{\mathbf{L}s^* \mathbf{R}(t_p)_* \mathcal{O}_{\mathrm{Spec}(\kappa(r_p))}\}_{p \in |S|} \rangle}.$$

For any $u \in |U|$, form the fibered square

$$\begin{array}{ccc} W_u := U \times_X \mathrm{Spec}(\kappa(u)) & \xrightarrow{b_u} & \mathrm{Spec}(\kappa(u)) \\ a_u \downarrow & & \downarrow t_u := s \circ h_u \\ U & \xrightarrow{s} & X. \end{array}$$

There is a canonical morphism $e_u: \mathrm{Spec}(\kappa(u)) \rightarrow W_u$ induced by the pair $(h_u, 1)$, and it satisfies $a_u \circ e_u = h_u$ and $b_u \circ e_u = 1_{\mathrm{Spec}(\kappa(u))}$.

By [Sta26, Tag 09TF], the morphism t_u is affine. Therefore, its base change a_u is affine. Since U is affine, it follows that W_u is affine. Moreover, b_u is the base change of the quasi-compact étale morphism s , so b_u is finitely presented. Thus, W_u is an affine Noetherian scheme. Flat base change gives

$$\mathbf{L}s^* \mathbf{R}(t_u)_* \mathcal{O}_{\mathrm{Spec}(\kappa(r_u))} \cong \mathbf{R}(a_u)_* \mathcal{O}_{W_u}.$$

In particular, (5.1) becomes

$$(5.2) \quad \mathbf{R}(a_r)_* \mathcal{O}_{W_r} \in \overline{\langle \{\mathbf{L}s^* \mathbf{R}(t_p)_* \mathcal{O}_{\mathrm{Spec}(\kappa(r_p))}\}_{p \in |S|} \rangle}.$$

Since W_r is affine, $D_{\mathrm{qc}}(W_r) = \overline{\langle \mathcal{O}_{W_r} \rangle}$, and so $\mathbf{R}(e_r)_* \mathcal{O}_{\mathrm{Spec}(\kappa(r))} \in \overline{\langle \mathcal{O}_{W_r} \rangle}$. Applying $\mathbf{R}(a_r)_*$ and Lemma 4.1 gives

$$\begin{aligned} \mathbf{R}(h_r)_* \mathcal{O}_{\mathrm{Spec}(\kappa(r))} &\cong \mathbf{R}(a_r)_* \mathbf{R}(e_r)_* \mathcal{O}_{\mathrm{Spec}(\kappa(r))} \\ &\in \overline{\langle \mathbf{R}(a_r)_* \mathcal{O}_{W_r} \rangle}. \end{aligned}$$

Combining this with (5.2), we obtain

$$(5.3) \quad \mathbf{R}(h_r)_* \mathcal{O}_{\mathrm{Spec}(\kappa(r))} \in \overline{\langle \{\mathbf{L}s^* \mathbf{R}(t_p)_* \mathcal{O}_{\mathrm{Spec}(\kappa(r_p))}\}_{p \in |S|} \rangle}.$$

We now describe the localizing subcategory on the right. For each $p \in S$, abbreviate $W_p := W_{r_p}$, $a_p := a_{r_p}$, and $b_p := b_{r_p}$. Flat base change gives

$$(5.4) \quad \mathbf{L}s^* \mathbf{R}(t_p)_* \mathcal{O}_{\mathrm{Spec}(\kappa(r_p))} \cong \mathbf{R}(a_p)_* \mathcal{O}_{W_p}.$$

For every $l \in |W_p|$, let $v_l: \mathrm{Spec}(\kappa(l)) \rightarrow W_p$ denote the natural morphism. Since W_p is an affine Noetherian scheme, [Neeg2, Theorem 2.8] and [Sta26, Tag 071Q] give

$$D_{\mathrm{qc}}(W_p) = \overline{\langle \{\mathbf{R}(v_l)_* \mathcal{O}_{\mathrm{Spec}(\kappa(l))}\}_{l \in |W_p|} \rangle}.$$

On the other hand, because W_p is affine, $D_{\mathrm{qc}}(W_p) = \overline{\langle \mathcal{O}_{W_p} \rangle}$. Applying $\mathbf{R}(a_p)_*$ and Lemma 4.1 in both directions therefore yields

$$(5.5) \quad \overline{\langle \mathbf{R}(a_p)_* \mathcal{O}_{W_p} \rangle} = \overline{\langle \{\mathbf{R}(a_p \circ v_l)_* \mathcal{O}_{\mathrm{Spec}(\kappa(l))}\}_{l \in |W_p|} \rangle}.$$

Define $\Sigma_p := a_p(|W_p|) \subseteq |U|$ and $\Sigma := \cup_{p \in S} \Sigma_p \subseteq |U|$. For every $l \in |W_p|$, the morphism $a_p \circ v_l: \text{Spec}(\kappa(l)) \rightarrow U$ represents the point $a_p(l) \in \Sigma_p$. Hence, after discarding repetitions and applying Lemma 4.2, equations (5.4) and (5.5) imply

$$(5.6) \quad \overline{\langle \{ \mathbf{L}s^* R(t_p)_* \mathcal{O}_{\text{Spec}(\kappa(r_p))} \}_{p \in |S|} \rangle} = \overline{\langle \{ \mathbf{R}(h_u)_* \mathcal{O}_{\text{Spec}(\kappa(u))} \}_{u \in \Sigma} \rangle}.$$

Notice also that $s \circ a_p = t_p \circ b_p$. Since t_p represents p , it follows that every point in Σ_p maps to p . Thus,

$$(5.7) \quad \Sigma_p \subseteq s^{-1}(\{p\}) \quad \text{and hence} \quad \Sigma \subseteq s^{-1}(S).$$

Combining (5.3) and (5.6), we obtain

$$\mathbf{R}(h_r)_* \mathcal{O}_{\text{Spec}(\kappa(r))} \in \overline{\langle \{ \mathbf{R}(h_u)_* \mathcal{O}_{\text{Spec}(\kappa(u))} \}_{u \in \Sigma} \rangle}.$$

The algebraic space U is a point-generated scheme. Therefore, [AJL24, Theorem 0.1] identifies its $\otimes$ -localizing subcategories with subsets of $|U|$. The preceding membership implies that $r \in \Sigma$. By (5.7), $r \in s^{-1}(S)$. Since $s(r) = q$, it follows that $q \in |S|$. Therefore, $\phi(\theta(S)) \subseteq S$. Together with the opposite inclusion proved above, this gives $\phi \circ \theta = 1$. We have now shown that ϕ and θ are mutually inverse. $\square$

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